

claude-windows / c6028e72-ebfd-4218-89a8-c4d7dd23be96

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c6028e72-ebfd-4218-89a8-c4d7dd23be96\subagents\workflows\wf_f9aa1d1c-c83\agent-abd593d3fafd23f9e.jsonl`  
- SHA-256: `1d47203cb71658611846e2d4a04087382437aca38c46e58491fc497deb6b1921`  
- Source modified: `2026-06-09T11:25:35+00:00`  
- Imported at: `2026-07-05T16:48:23+00:00`  
- project: `wf_f9aa1d1c-c83`  
- session_id: `c6028e72-ebfd-4218-89a8-c4d7dd23be96`
```

Transcript

1. user (2026-06-09T11:24:27.726Z)

Restate, crisply, the owner's reframed vision + constraints below as a requirements brief for the strategic decisions (model strategy, conversational-QA, multisport, language). No web needed.

OWNER'S REFRAMED VISION (from PR #61 review comments ? supersedes earlier framing where they differ):

- IMMEDIATE GOAL: a TRAINING FRAMEWORK + HARNESS to produce a VERY HIGH precision/recall model that extracts

- HIGHLIGHTS and RALLIES from a clip. Quality first. On-device is LATER (a model-compression goal for power users to

- save upload cost) ? NOT the immediate goal; don't over-index on it now.

- MULTISPORT FROM THE START: all NET-SEPARATED RACQUET SPORTS (badminton, tennis, table tennis, pickleball, padel)

- first, eventually ALL sports. A SINGLE, sports-AGNOSTIC model (the collector + indexer are already sport-generic by

- design; the repo will be renamed off "badminton"). More labels/compute/storage is an accepted "good problem."

- CONVERSATIONAL VIDEO-QA is a north-star feature: upload a video -> the model processes it -> the user asks CONTEXTUAL

- QUESTIONS continuously, e.g. "show me the longest rally", "make a video of all the service faults in this clip",

- "show me a clip where the rally lasted over 20 shots". This implies per-video

- STATE/bookkeeping the model itself

- won't hold ? so the FRAMEWORK must be designed for that bookkeeping from the start. Owner asks: feasible to embark on

- now, or punt and revisit after quality is solid?

- HARD COMMERCIAL-LICENSE GUARDRAIL (owner: "never corrupted"): EVERY dependency ? code, data, AND model weights ? must

- be usable in PRIVATE PROPRIETARY COMMERCIAL software. MIT / Apache-2.0 / BSD are the easy fits. Reject viral/NC/

- custom-restrictive (e.g. Gemma-3 viral terms; weights trained on paid-LLM outputs; >X-MAU caps if truly binding).

- NEVER train on paid-LLM (Gemini/OpenAI/Anthropic) outputs. Exclude minors; separate training opt-in. Hopefully a

- CONVERSATIONAL single model.

- CURRENT STATE (de-risking evidence): on a 6-match human-labeled corpus, a learned per-frame prototype gets held-out

- AUC 0.83-0.92 and a cross-video LOVO mean 0.583 vs the heuristic's 0.480 ? and the learned model DECOUPLES from the

- multi-court "background games" confound the heuristic can't escape. WASB = the shuttle detector (MIT code, but the

- distributed checkpoint is academic-data-trained ? provenance unresolved = a commercial-cleanliness blocker to fix).

- CONSTRAINTS: solo founder, Bangalore; local dev box = ONE RTX 2070 Super (8GB) + WSL2; paid

Gemini stays the fallback;
cloud GPU rented per-job (Modal/RunPod per the earlier study).

2. assistant (2026-06-09T11:24:32.560Z)

I'll restate the owner's reframed vision as a requirements brief. Let me first check the PR #61 context and the de-risking evidence to ground the brief accurately.

3. user (2026-06-09T11:24:34.586Z)

```
{"body":"Single consolidated PR for offline review ? the full quality iteration after golden labels arrived. **Docs + eval-tooling + a prototype only; no production pipeline change. Owned-model = design/plan only (no implementation).**\n\n## TL;DR of the findings\n- Built a **5-match human-labeled corpus** (2 singles, 1 doubles, 2 multi-court; 3 cameras/frame-rates) + honest cross-video **LOVO**.\n- **The learned prototype overtakes the heuristic** and scales with data: learned LOVO mean 0.377 (n=2) ? 0.615 (n=3) ? **0.568 (n=5)**; heuristic **0.544 ? 0.463** (it *falls* as the corpus gets more representative). Gap inverted (heuristic led at n=2; learned +0.105 at n=5).\n- **Root cause of the heuristic's failures = MULTI-COURT (background games).** WASB tracks every shuttle incl. other courts ? dead-time shuttle visibility 41?57% (vs 18?22% clean) ? boundaries bleed. The **contrast** metric (in-rally÷dead-time vis) predicts success (?3.6× works, ?1.9× fails). **Academies are multi-court = the target env**, so these are the most representative videos. The learned model recovers them (AUC 0.83; Kushagra 0.561, gBaaddy 0.574 vs heuristic 0.16/0.28).\n\n## What's in here\n- **`backend/eval/rally_seq_proto.py`** (+3 tests) ? learned per-frame rally segmenter (owned-model **Gen-0 seed**): 6 hand-built WASB-trajectory features ? dependency-free LogisticRegression, honest LOVO. Held-out AUC 0.83?0.92.\n- **`docs/archives/quality-iterations/2026-06-multivideo-lovo/`** ? the full learnings: n=2?n=5 LOVO, the crossover, the multi-court root cause + contrast metric, and a **heuristic-discovery** section (foreground spatial gate = net ?0.04, continuity filter = no-op; **new lead: AUDIO for foreground-vs-background discrimination** ? the mic sits at the foreground court, so foreground hits are loud/near, background quiet/distant).\n- **`docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`** ? 11-agent verified study (35 sources): 3 generations off one corpus (Gen-0 ASFormer/MIT ? Gen-1 fusion ? Gen-2 Qwen3-VL-4B/Gemma-4, Apache-2.0; Gemma-3 rejected); platforms (Modal/RunPod/Vertex through the privacy+cost lens); label-reuse; guardrails; costs; critic caveats (WASB checkpoint provenance C3, purge Gemini-trained labels).\n- **`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`** ? phased plan (Gen-0 first, gated on n?5 + scaffolding + your sign-off).\n- **`docs/CODE_MAP.md`** ? added the new eval modules + the multi-court gotcha.\n\n## 4 decisions for you (in the plan)\n1. Which Phase-0 milestone to greenlight first (I recommend the **corpus spine**). 2. WASB C3 provenance stance. 3. Ratify `Apache-2.0/MIT`. 4. Confirm the collector is the corpus system-of-record.\n\n## Honest caveats\n\nPrototype is **directional** (n?5, linear, speed feats not fps-normalized, threshold-transfer unsolved ? real ship gate is a segment-level A/B at n?5 on the GPU box). Heuristic-discovery filters were **negative** (reported as such). TestLargeVideo (6 rallies) is too short to weight.\n\nTests: indexer **254 green**. Prior merged this iteration: #60 (partial-label harness), collector #13 (ingestion guard).\n\n? Generated with [Claude Code](https://claude.com/claude-code)","comments":[{"id":"IC_kwDOSg1-f88AAAABFT4Jqw","author":{"login":"avidullu"},"authorAssociation":"OWNER","body":"Update: prototyped the audio-foreground idea this session (no new videos needed ? used the 2 multi-court labelled videos + a clean control). **Honest result: weak.** Audio onset/RMS energy AUC 0.53?0.65 for in-rally vs dead-time; HF bands (8?22 kHz) improve contrast (distance-attenuation physics is real) but AUC stays ~0.6. It fails because the foreground court is loud during dead-time too (footwork/talk/pickup + gym reverb), so audio loudness ? court occupancy, not rally. Doc corrected ? audio is no longer flagged as the top lead. The robust multi-court path stays the learned visual model (AUC 0.85); the remaining heuristic lever is proper court-region detection.", "createdAt":"2026-06-08T16:56:02Z","includesCreatedEdit":false,"isMinimized":false,"minimizedReason":"","reactionGroups":[],"url":"https://github.com/avidullu/badminton-highlight-indexer/pull/61#issuecomment-4651354539","viewerDidAuthor":true},{id":"IC_kwDOSg1-f88AAAABFT6aoA","author":{"login":"avidullu"},"authorAssociation":"OWNER","body":"Follow-up on court-region detection (the remaining heuristic lead): **smoke-tested, low headroom.** A feasibility probe (numpy, no model) shows only **12?14%** of dead-time shuttle detections on the multi-court videos fall *outside* the foreground rally region ? so 86?88% of the confusion is *inside* the foreground court. A perfect court gate would lift contrast only to ~1.6?2.2× (still failing). The confusion is the foreground court's own between-point activity, not separable background courts. **So court detection (heuristic or model) won't meaningfully help.** Net: 3 heuristics tested + failed this session (spatial gate, audio, court separability) ? all because
```

it's a *temporal* rally-vs-non-rally problem in the same space, which only the learned model solves (AUC 0.83). No cheap shortcut; the learned segmenter is the path.", "createdAt": "2026-06-08T17:00:45Z", "includesCreatedEdit": false, "isMinimized": false, "minimizedReason": "", "reactionGroups": [], "url": "https://github.com/avidullu/badminton-highlight-indexer/pull/61#issuecomment-4651391648", "viewerDidAuthor": true}], "title": "Quality iteration: human-GT multi-video LOVO + learned prototype + multi-court finding + owned-model study/plan"}

4. user (2026-06-09T11:24:34.796Z)

BACKLOG.md
CODE_MAP.md
COMMERCIALIZATION.md
DEFERRED_HARDENING_PLAN.md
EVALUATION.md
GOLDEN_SET_IMPLEMENTATION.md
GOLDEN_SET_PHASE1_VIDEOS.md
GOLDEN_SET_VENDORING.md
HOW_RALLY_DETECTION_WORKS.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PLATFORM_ARCHITECTURE.md
PMF_MARKET_RESEARCH.md
README.md
ROADMAP.md
TRACKNET_WSL_SETUP.md
VIDEO_RECORDING_GUIDELINES.md
archives
---ARCHIVES---
COMMERCIAL_READINESS_REVIEW.md
MONETIZATION_AUDIT.md
README.md
checkpoints
decisions
past_projects
quality-iterations
research

5. assistant (2026-06-09T11:24:38.927Z)

The PR body and owner comments confirm the de-risking evidence. Let me quickly check the owned-model plan and study to ground the model-strategy and conversational-QA pieces precisely.

6. user (2026-06-09T11:24:39.554Z)

```
1      # Owned-model implementation plan (DRAFT ? for owner review, no code started)
2
3      **Status:** PLAN ONLY. No owned-model implementation begins until the owner greenlights
a specific
4      milestone (per CLAUDE.md). Paid Gemini stays the always-ready primary/fallback
throughout.
5      **Grounded in:** `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` (the verified
study) +
6      `docs/archives/quality-iterations/2026-06-multivideo-lovo/` (the LOVO + prototype that
de-risk it).
7
8      ## North star & non-negotiables
9      - **Goal:** an OWNED, on-device rally?highlight model; local inference = the privacy
moat. The moat is the
10     consented dataset + eval harness, **not** the weights.
11     - **Guardrails (hard):** Apache-2.0/MIT bases only (Gemma-3 rejected); **never** train
on paid-LLM
12     (Gemini/OpenAI/Anthropic) outputs; exclude minors + separate training opt-in; split
data **by match**.
13     - **Sequencing discipline:** committed next platform slice is the **async job model
(scaffolding)**. Heavy
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14     model *training* (Gen-2, cloud) waits for P0?P4 scaffolding + a healthy corpus +
explicit go. The
15     **corpus + data + compliance foundation is no-regret and can proceed first** ? it's
what makes later
16     training cheap and clean.
17
18     ## Why now is the right moment (today's evidence)
19     - Heuristic cross-video LOVO =  $0.443 \pm 0.279$  (Adarsh 0.722, Kushagra 0.163-ceiling).
The  $\pm 0.279$  is
20     *method* variance ? the unsupervised windowing is footage-fragile even when WASB
recall is excellent
21     (Kushagra: 97% of rallies have shuttle motion, yet unsegmentable).
22     - The learned prototype gets held-out per-frame AUC 0.84?0.88; Kushagra oracle F1
0.578 vs 0.163.
23     ? The rally signal is learnable; the bottleneck is data scale + calibration (n?5),
not method**.
24
25     ---
26
27     ## Phase 0 ? Foundation (start now; no GPU/scaffolding dependency)
28
29     ### M0.1 ? The labeled-corpus spine (the moat) . *owner-greenlight item*
30     - What: one canonical JSONL row per rally: `{video_id (file MD5), rally_ordinal,
sport, start, end,
31     ending_reason, label_type, source(human|pseudo|vendor), annotator/provenance,
consent/training_opt_in,
32     split(train|val|test, assigned BY MATCH), trajectory_ref, labeled_window}`.
33     - Where: authored in `sports-data-collector` (system-of-record; extends PR #13 + the
positioning doc);
34     the indexer only consumes transcoder outputs.
35     - 3 deterministic transcoders (one row ? three views, zero re-labeling): (1)
heuristic/LogReg
36     candidate labels (already exists: `training.label_candidates`), (2) TAL ActivityNet-
JSON, (3) VLM clip?QA.
37     - Guardrail enforcement: provenance/consent/minors fields gate corpus membership
before transcoding;
38     Gemini labels never enter a training transcoder; the new `eval_partial_labels` window
flows into every view.
39     - One scorer spine: `metrics.match_intervals` for heuristic-labeling, TAL eval, and
the Gen-2 reward.
40     - Acceptance: round-trip test (collector export ? each transcoder ? indexer
loaders); split-by-match
41     enforced; no Gemini-sourced rows in any training view.
42
43     ### M0.2 ? Grow n?2 ? 5 distinct matches . *owner action + my scoring loop*
44     - What: tag ?3 more distinct matches (gBaaddy, TestLarge, + a small Rooru batch).
I generate the
45     trajectory + score each as it lands (the loop we've been running).
46     - Healthy corpus bar: ?5 distinct match-sessions with full human labels ? stable
leave-one-match-out.
47     - Acceptance: n?5; per-match + LOVO numbers recorded; corpus diversity
(camera/court/lighting) noted.
48
49     ### M0.3 ? Compliance cleanup (do before it bakes in) . *owner-greenlight item*
50     - (i) Purge Gemini-derived TRAINING labels. The distilled pipeline currently trains
on Gemini silver
51     (`training.label_candidates` over Gemini CSVs) ? this violates the no-paid-LLM-
training rule. Retarget
52     training to human + open-teacher (WASB/own-model) labels only; Gemini stays
eval/teacher-reference.
53     - (ii) WASB checkpoint provenance (C3). The distributed
`wasb_badminton_best.pth.tar` is academic-data-
54     trained (provenance unresolved) ? it propagates to anything trained on its features.
Decide: document
55     as a hard Gen-0 prerequisite and plan ROADMAP Phase-4 "train own WASB weights," or

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scope an alternative.
56     - **Acceptance:** no paid-LLM outputs in any training path; a written C3 decision +
prerequisite flag.
57
58     ### M0.4 ? Gen-0 A/B harness (built here; trains on your GPU box) · *owner-greenlight
item*
59     - **What:** productionize the prototype into a real harness ? feature stack (visibility-
run-length, velocity,
60     acceleration, net-crossing flags, smoothing/windowing) ? **ActionFormer (MIT, 1:1
rally=instance)** and/or
61     **ASFormer (MIT, TAS)** train?eval?**ship-gate**. Reuse `metrics.match_intervals`.
62     - **Ship gate (the honest bar):** held-out per-match F1 must beat the **honest cross-
video heuristic LOVO
63     ( $\sim 0.44$ )**, not the fit-on-self 0.73. nanochat-style pass/fail verdict.
64     - **Where it runs:** training/eval on the **GPU/WSL machine** (can't run in this dev
container). Harness +
65     feature stack are authored here; you run the A/B.
66     - **Acceptance:** one-command train?eval?verdict; reproducible; the prototype's AUC
translated to segment F1
67     at n?5; documented A/B (ActionFormer vs ASFormer vs heuristic).
68
69     ---
70
71     ## Phase 1 ? Gen-0 ship decision (gated on n?5 + M0.4)
72     - Run the A/B; if a learned Gen-0 clears the honest LOVO bar with per-video calibration,
**it becomes the
73     rally segmenter** (still fully local, MIT, on the 2070 Super). If not, the finding
(more data / better
74     features needed) is itself the next-step signal ? no over-fitting to declare victory.
75     - **Decision gate:** Gen-0 adopted ? more data/features first. Either way, the corpus +
harness persist.
76
77     ## Phase 2 ? Gen-1 / Gen-2 (gated on platform scaffolding P0?P4 + explicit go)
78     - **Gen-1 (Option A):** concatenate fully-local cue channels (audio-onset, pose/court)
onto the same head;
79     fusion in our rally layer. License/cost-vet the pose extractor (open item).
80     - **Gen-2 (Option B):** e2e multimodal VLM on an Apache-2.0 base (**Qwen3-VL-4B**
primary / Qwen2.5-VL-7B
81     fallback / **Gemma 4** end-state once a timestamp recipe is proven), SFT cold-start ?
GRPO/RFT with a
82     temporal-IoU-vs-golden reward. **Cloud training on Modal** (best no-access contract);
base license
83     re-verified per-checkpoint.
84     - **Hard pre-serving gate:** the **$/call crossover benchmark** (self-hosted-VLM vs
Gemini on real workload)
85     + an **on-device latency/throughput** measurement of the quantized base on the 2070
Super ? both currently
86     *unmeasured* and load-bearing. Don't stand up an owned serving stack until these clear
(COMMERCIALIZATION C7).
87
88     ---
89
90     ## Decisions I need from you (to unblock specific milestones)
91     1. **Greenlight which Phase-0 milestone(s) to start in code** (M0.1 corpus spine = my
recommendation; M0.3
92     cleanup; M0.4 harness). M0.2 (tagging) runs regardless.
93     2. **WASB C3 stance** ? accept "document as prerequisite" for now, or prioritize own-
WASB-weights sooner?
94     3. **Ratify "Apache-2.0/MIT"*** (the study's approved list includes MIT bases/teachers;
the guardrail text
95     says "Apache-2.0 only").
96     4. **Corpus home** ? confirm the collector is the system-of-record for the spine
(consistent with PR #13).
97
98     ## Open prerequisites (carried from the study's critic ? none block Phase 0)

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99 WASB C3 provenance · Gemma-4 license stays AMBER until a raw HF LICENSE check · on-
device VLM latency
100 unbenchmarked · real VLM VRAM/token cost on our 1080p/4K-60fps clips · DPA + no-train
for any cloud training ·
101 the owned-vs-Gemini \$/call crossover volume.
102
103 ---
104 *This plan is design-only and reversible. Nothing here starts until you greenlight a
milestone.*
105

7. assistant (2026-06-09T11:24:43.876Z)

I have enough grounding from the PR body, owner comments, and the implementation plan. The owner's reframed vision in the prompt is the authoritative source where it differs from earlier framing (notably: on-device demoted from immediate goal to a later compression goal; multisport from the start; conversational QA as north-star). Let me produce the requirements brief.

8. user (2026-06-09T11:25:24.400Z)

Structured output provided successfully

9. assistant (2026-06-09T11:25:35.237Z)

Requirements brief delivered via StructuredOutput. Summary of what I grounded it on and the key reframes captured:

Sources read: PR #61 body + the two owner comments (audio-foreground and court-region negative results), and `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`.

Key reframes the brief encodes (where the new vision supersedes earlier framing):

- On-device demoted from "immediate privacy moat" to a *later* compression goal ? the immediate target is the training framework/harness and model quality.
- Multisport (all net-separated racquet sports) from day one, via a *single sport-agnostic model* + repo rename off "badminton".
- Conversational video-QA elevated to north-star, with per-video state/bookkeeping as a *framework* requirement from the start ? plus the explicit open question (embark now vs punt).

16 requirements / 8 tensions returned. The sharpest tensions worth the owner's attention: multisport-from-start vs. a badminton-only evidence base (WASB features won't obviously generalize); single conversational model vs. the no-paid-LLM-training guardrail (no Gemini teacher allowed); and the quality bar ("very high precision/recall") vs. the honest current ceiling (~0.583 LOVO, n=6, directional).

Relevant file paths (all absolute):

- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED\_MODEL\_TRAINING\_STUDY.md`
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\quality-iterations\2026-06-multivideo-lovo\`